Tuscany Pyroconvection-Type Forecast – ICON-EU model levels + ICON-2I 2.2 km gate – VALID 2026-07-16

3-hourly, 24 h. Run 2026-07-16 00Z. Method after Castellnou et al. (2022), JGR-Atmos.

How to read this product

Two panels are produced. The **fuel-gated** map is the expected, operationally comparable product (the equivalent of the Catalan “tipus de piroconveccio” map): a pyroCu/pyroCb class is assigned **only where a fire could actually reach >= 10 MW/m** of fireline intensity on the real Tuscany fuels. The **potential** map is an unconditional **upper bound** – it assumes a pyroCu-capable fire in *every* cell. Use the gated panel for situational awareness; use the potential panel only to see the atmospheric ceiling.

Method and its limits (read this before using the classes)

The column is classified from a **real vertical profile**, not a standard atmosphere: per-cell heights from the ICON-EU model-level heights (HHL), the mixing depth from the **bulk Richardson number** (first Rib >= 0.33 above 200 m AGL, referenced to the 2 m / 10 m state), the LCL from the exact **Bolton (1980)** formula, and the humidity at the ABL top from the model's RH. The cap gamma-theta is taken over ABL+200 m to ABL+1200 m.

The **mixed-layer stability** here is a genuine measurement, not a proxy. The ICON-EU native model levels put ~10 levels inside the mixed layer (this run: 1 median), so the mixed-layer dtheta/dz is a real least-squares fit across those levels. Validated against IGRA radiosondes (JJA 12Z, period of record), the model-level fit cuts the gradient bias to ~-0.4e-4 K/m (from ~-4.7e-4 on ICON-2I's 5 pressure levels) and the Rib ABL bias to ~-70 m (from ~-700 m). This is why the hybrid product exists: the ICON-2I 2.2 km open data cannot resolve the mixed layer, and ICON-EU can.

The trade is horizontal resolution: the atmosphere is 6.5 km (regridded to the 2.2 km grid), while the **fuel gate keeps ICON-2I's 2.2 km surface fields**, where fine terrain matters. The mixed-layer gradient is now measured, but the 1.1e-3 K/m threshold still sits inside the validated error bar (+/- ~2.6e-4), so treat class *counts* as indicative, not exact.

Ladder actually used for this run: `shear`. The full method has a fifth diagnostic – the distance from the ABL/LCL to the height of maximum wind shear – which this product does not yet compute, so the four-diagnostic ladder runs. The shear clause is a *necessary* condition for the top class, so omitting it makes class 4 somewhat **easier** to reach here than in the full method. Treat class 4 as an alert to inspect the column, not as a calibrated probability.

The classes express atmospheric predisposition **given a fire of sufficient power**; in the gated panel that power is computed, not assumed. They are paper-informed thresholds, not locally validated ones.

Expected pyroconvection type – FUEL-GATED

Atmospheric potential – UPPER BOUND (assumes a pyroCu-capable fire in every cell)

Class scale (low -> high pyroconvective activity)

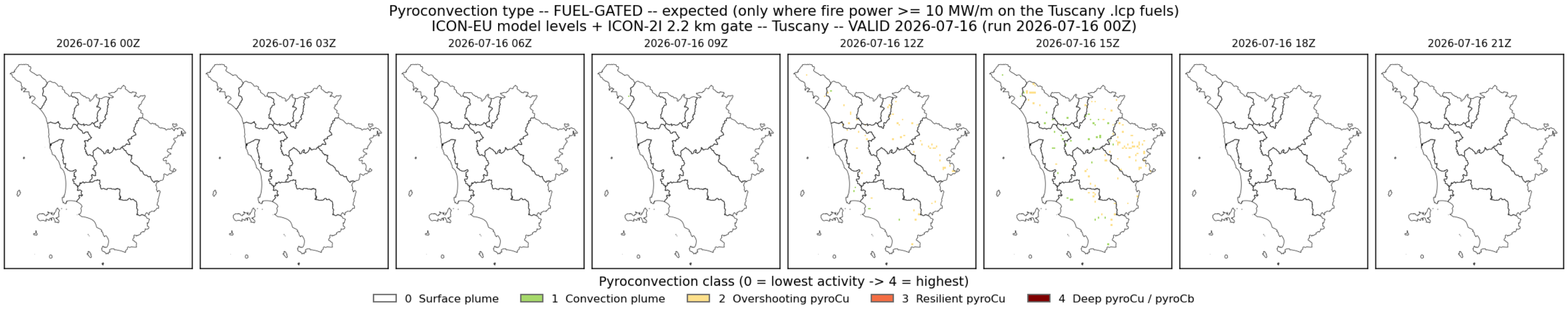

Figure 1: gated

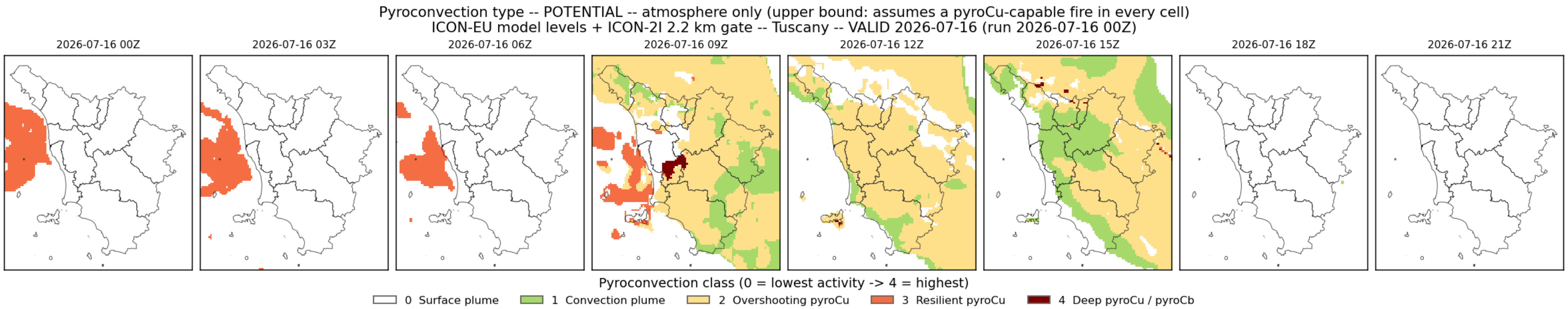

Figure 2: potential

Level	Colour	Class	Meaning
0	white	Surface plume	Buoyant smoke plume; no significant cloud development.
1	green	Convection plume	Plume penetrates a stable mixed layer; condensation possible, no pyroCu.
2	yellow	Overshooting pyroCu	Brief pyrocumulus; cloud base above the mixing height (LCL/ABL > 1).
3	orange	Resilient pyroCu	Persistent pyrocumulus in an unstable column (LCL/ABL < 1).
4	dark red	Deep pyroCu / pyroCb	Deep pyroconvection / pyrocumulonimbus; weak upper cap lets the plume deepen.

Classification thresholds (ladder in use: `shear`)

Diagnostic	Threshold	Effect
ML dtheta/dz (least-squares fit, in mixed layer)	> 1.1e-3 K/m (stable)	Convection plume only – no pyroCu
ML dtheta/dz (measured on model levels)	<= 1.1e-3 K/m	Column is pyroCu-capable (bias ~-0.4e-4 K/m vs radiosondes)
LCL / ABL ratio	1.0 – 1.60	Overshooting pyroCu (brief)
LCL / ABL ratio	< 1.0	Resilient pyroCu (persistent)
LCL / ABL ratio	<= 1.10 (+ conditions below)	Admissible for deep pyroCu / pyroCb
Cap gamma-theta (ABL+200 m -> ABL+1200 m)	<= 4.2e-3 K/m (weak cap)	Permits deepening to pyroCb
Cap gamma-theta	>= 4.8e-3 K/m (strong cap)	Inhibits deepening (resilient at most)
RH at the ABL top (mean, ABL +/- 150 m)	>= 60%	Required for classes 3 and 4 (was 80%; relaxed for dry fire weather)
Shear-maximum distance / ABL	<= 0.30	Required for class 4 **in the 5-diagnostic ladder only** (not resolvable here – see Method)
Fireline intensity (fuel gate, gated panel)	>= 10 MW/m	Minimum fire power for any pyroCu (Tedim et al. 2018)
ABL depth	< 600 m	Held at surface plume (mixing too shallow)
Usable pressure levels	< 4	Cell not classified

Reference cases (Castellnou et al. 2022, Table 1)

The paper's labelled events, which anchor the published 3-diagnostic ladder (`pyflam.pyroconvection_type(ladder="castellnou")`, still the library default and regression-tested against these rows). The profile ladders used for this map are stricter at the top: they additionally require a moist ABL top and an LCL close to it.

Case	Observed type	LCL/ABL	ML dtheta/dz	gamma-theta (700-500)
T21	Convection plume	–	stable	–
SCQ32	Overshooting pyroCu	> 1	neutral/unstable	–

Case	Observed type	LCL/ABL	ML dtheta/dz	gamma-theta (700-500)
M11	Resilient pyroCu	< 1	unstable	4.2e-3 (resilient cap)
SCQ41	pyroCu, not pyroCb	< 1	unstable	5.1e-3 (strong cap)
SCQ51	Deep pyroCu / pyroCb	< 1	unstable	3.9e-3 (weak cap)

Forcing: atmosphere from ICON-EU 6.5 km native model levels (DWD open data, CC-BY), lowest ~24 levels; ~1 inside the mixed layer here, regridded to the ICON-2I 2.2 km grid. Surface fields and fuel gate from ICON-2I 2.2 km (MISTRAL / AgenziaItaliaMeteo). Classifier: pyflam.pyroconvection_type (bulk-Richardson ABL, Bolton LCL, measured mixed-layer dtheta/dz, cap gamma-theta, ABL-top RH; no surface CAPE). Gate: Rothermel + Cruz-2005 crown on the .lcp fuels with forecast moisture/wind. Per-hour diagnostic rasters (ABL, LCL, LCL/ABL, ML dtheta/dz, cap, RH-top) accompany the classes. Province borders: ISTAT-derived (openpolis geojson-italy). Generated by tests/pyroconv_daily.py.